

DAYANARA LEBRON-ALDEA, M.S.

Data Scientist | Bioinformatics & Statistical Leadership | Product, QC & Team Management | Educator

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PROFESSIONAL SUMMARY

Data Scientist and technical leader with 10+ years of experience spanning applied biostatistics, machine learning, production bioinformatics pipelines, quality systems, and cross-functional product delivery. Has independently led a data product from R&D through a \$1M biopharma contract, built and directed a Quality Control function overseeing a team of 4 scientists, and repeatedly been told by managers and peers that she is ready for people-management and senior leadership roles — feedback she has spent the years since deliberately building the technical depth to fully back up. Co-inventor on a published U.S. patent application, author of three peer-reviewed publications, and 5+ poster presentation awards in life sciences and data science. A confident, clear communicator with a long parallel record as an educator: has taught or mentored 250+ students across university, corporate, and independent settings, starting as a biostatistics instructor at age 19–20. Bilingual (English/Spanish). Currently completing a deliberate pivot toward technical product and program leadership, supported by a Google Project Management Professional Certificate and a competitively awarded Leland Product Management program grant.

LEADERSHIP & MANAGEMENT HIGHLIGHTS

- Led end-to-end delivery of the eSHAPE data product at Eclipse Bioinnovations — from pipeline R&D through commercialization — generating a \$1M biopharma contract within months of launch and contributing to a published U.S. patent application.
- Founded and directed Eclipse Bioinnovations' Quality Control department from the ground up: designed the pipeline, experimental design, and sample-size methodology, and directly oversaw a team of 4 scientists.
- Served as Statistical Lead managing all statistical applications across customer-facing projects at Eclipse, expanding the company's service offerings and contributing a \$300K+ increase in first-year earnings.
- Oversaw and mentored bioinformatics engineers (2 at Eclipse; supervised a broader engineering team on pipeline infrastructure at Tecan), directly managing their project assignments and output.
- Led recruitment, hiring, and onboarding of bioinformatics scientists at Tecan Genomics: wrote job descriptions, conducted interviews, and managed 2 new hires through onboarding and ramp-up.
- Acted as Product Lead at Eclipse: owned multi-phase, cross-functional projects, delegated tasks, managed onboarding and training-material creation, and served as point of contact for resolving customer escalations.
- Cross-collaborated with Sales, Marketing, R&D, and Software Engineering teams at Tecan and Eclipse to translate technical capability into customer-facing products and go-to-market materials.
- Built and led an internal Data Quality/QC training and onboarding program, authoring SOPs adopted company-wide for analytical protocols and performance benchmarks.
- Has independently run her own consulting and education practice since November 2023 — the same ownership, client management, and delivery accountability as a small business lead.

SKILLS

Statistical Methods: Logistic regression, mixed-effects models, PCA/factor analysis, Bayesian inference, survival analysis, bootstrap, hypothesis testing, DESeq2 differential analysis, G-BLUP models

Programming & Pipelines: Python (pandas, NumPy, SciPy, scikit-learn, PyTorch, TensorFlow), R, SQL, HTML, Bash, Snakemake; production pipelines on AWS (EC2, S3) and GCP; HPC/SLURM

Bioinformatics: RNA-Seq, scRNA-Seq, DNA-Seq, WES, Methyl-Seq, eCLIP, miRNA-Seq, SHAPE-Map, Variant Calling, Metagenomics; STAR, Bowtie2, samtools, bcftools, RNAFold/RNAstructure, DESeq2, EdgeR; NCBI, TCGA, ENCODE

Databases: SQL (PostgreSQL), NoSQL/MongoDB, relational database design, large-scale genomic data queries

Visualization & Reporting: Plotly, Seaborn, Matplotlib, Tableau, dynamic HTML reporting, technical documentation for mixed technical/non-technical audiences

Product, Program & QC Leadership: Product lifecycle management, cross-functional team leadership, hiring & onboarding, SOP authorship, QC framework design, client-facing delivery, budget/timeline ownership

Tools: Jira, Confluence, Smartsheets, Git/GitHub, Docker, Jupyter, Excel, MS Office (Word, Excel, PowerPoint)

Languages: English and Spanish (bilingual, professional)

WORK EXPERIENCE

Founder & Principal Consultant

Nov 2023 – Present

Independent Practice — “Yiyi’s Little Corner”, Puerto Rico (Remote)

- Built and operate an independent education and analytics consulting practice, serving 12+ recurring students/clients per year across mathematics (K–12), programming, and Spanish-language instruction — owning acquisition, curriculum design, scheduling, and delivery end to end.
- Ran an e-commerce resale storefront under the Yiyi’s Little Corner name: sold ~1,000 items in two years across Mercari, eBay, Poshmark and Facebook Marketplace while maintaining a 5.0-star seller rating, managing inventory, pricing, listings and customer service end to end.
- Completed the Google Project Management Professional Certificate (2024) and a competitively awarded Leland Product Management program grant (2026), building formal product and program management skills alongside continued technical consulting work.

Senior Genomics Data Scientist / Product Lead

Aug 2021 – Nov 2023

Eclipse Bioinnovations Inc., San Diego, CA (Remote)

Technical Delivery

- Developed an end-to-end pipeline predicting mRNA secondary structure transcriptome-wide from cell lines in in-vitro and in-vivo settings, using ML and bioinformatics tools in Python, achieving ~90%+ predictive accuracy across experimental datasets.
- Designed and executed validation studies correlating ML-derived RNA structure metrics against experimental outcomes across multiple cell lines and experimental conditions.
- Led pipeline automation using Snakemake in AWS EC2 with output to S3 (and Google Cloud), decreasing turnaround time for customer and routine data analysis by more than 60%.
- Implemented differential expression testing (DESeq2, mixed-effects negative binomial) to characterize RNA-protein interactions, incorporating batch correction and experimental variability, for both custom and internal R&D projects.
- Enhanced customer-facing HTML reports with dynamic visualizations of experimental and statistical results via Plotly and Seaborn.

Product & Business Leadership

- Led end-to-end product lifecycle for the eSHAPE platform, contributing to a \$1M biopharma contract for vaccine development and to a published U.S. patent application for prediction of secondary structure of LNP-coated RNAs.
- Served as Statistical Lead managing projects of statistical applications to customer datasets, expanding the company's services and yielding a \$300K+ increase in first-year earnings.
- Acted as Product Lead overseeing multi-phase projects with cross-functional teams; managed onboarding procedures, created training materials, delegated team tasks, and led hiring of talent.
- Managed customer projects end to end — tracking timelines and budgets and ensuring customer satisfaction at each step; acted as point of contact to resolve customer concerns and project escalations.

Quality Control & Team Leadership

- Spearheaded creation of the Quality Control department at Eclipse: directed pipeline development, experimental design, sample-size estimation, and statistical analysis for eCLIP (RBP-protein interaction) validation. Oversaw a team of 4 scientists.
- Authored SOPs covering process workflows, software standards, analytical protocols, and performance benchmarks, adopted company-wide.
- Provided training presentations on technical topics in data simulation, data management, and statistical applications to scientists across R&D and Bioinformatics departments.
- Provided direct customer support for technical questions and troubleshooting, including project-timeline reporting and secure data delivery via SFTP.

Senior Bioinformatics Data Scientist

Jun 2019 – Aug 2021

Tecan Genomics Inc., Redwood City, CA (Hybrid)

- Served as Product Analytical Lead for NGS Library Prep (DNA-Seq and RNA-Seq protocols), directing pipeline development in Python, maintenance, experiment analysis, and assay validation.
- Served as statistical expert providing guidance on appropriate statistics for experimental design, sample comparisons, and application of statistical methods across customer projects.
- Designed and maintained production-grade pipelines for sequencing data processing and quality assessment in GCP; developed QC frameworks and performance metrics evaluating assay robustness and reproducibility.
- Led recruitment of bioinformatics scientists — writing job descriptions, conducting interviews, and managing 2 new hires — ensuring team alignment with project goals.

- Led and supervised a team of software engineers in building a customer analysis portal and restructuring pipeline infrastructure in GCP toward containerized deployment.
- Developed onboarding materials and provided training to bioinformatics scientists and field application scientists on products, analysis pipelines, and standard procedures.
- Cross-collaborated with Sales and Marketing to develop marketing collateral demonstrating product applications in the field.
- Created targeted genotyping designs, managing bioinformatics projects and reporting analysis outcomes directly to customers.

Data Scientist / Information Systems Analyst

Jun 2015 – Apr 2019

Global Security, Lawrence Livermore National Laboratory, Livermore, CA

- Completed an M.S. in Statistics (UC Davis) concurrently while working full time in this role.
- Progressed from Computation Intern (CO-OP, Jun 2015) to Post-College Appointee in Information Systems (Feb 2016) to Data Scientist, under the mentorship of Dr. Jonathan Allen (Global Security, Bioinformatics Group).
- Developed and executed QC analysis on 300,000+ bacterial, viral, and fungal genomes extracted from NCBI, creating a tailored, genome-dense relational database for a metagenomics pipeline (LMAT).
- Compiled a metagenomics community profile of contaminants from 1000 Genomes Project samples and created a contaminant database used in QC of other metagenomic experiments — work that won Best Poster Presentation in Computation at the LLNL Symposium.
- Built the genome QC pipeline used before adding sequences to the LMAT database (R, Python, Bash, Perl on LLNL HPC clusters Catalyst and Herd): compared candidates against existing genomes, extracted contaminant fragments, and screened against human genomes with BLAST. This work surfaced an unexpected contamination finding, presented at the 2016 LLNL Poster Symposium.
- Developed a pipeline that produces BIOM tables to simplify interpretation of genome taxonomic profiles (R: phyloseq, ggplot2), and ran the Human Genome Project contaminant analysis across sequencing centers.
- Collaborated directly with the FDA and NASA to develop comprehensive metagenomic databases, deploying analyses and reporting results to federal stakeholders; co-developed the FDA-ARGOS genome database for infectious disease diagnostics, published in Nature Communications (Sichtig et al., 2018).
- Analyzed TCGA RNA-Seq datasets using an in-house metagenomic toolkit and Phyloseq, identifying associations between cancer and viral/bacterial presence in subjects.
- Programmed Python parsing scripts to mine external sources and compile an antimicrobial resistance SQL/Postgres database.
- Processed and launched LMAT analysis requests from external collaborators and extracted actionable information from its taxonomic outputs; contributed to an external xenosurveillance project detecting malaria in mosquitoes in Africa.
- Mentored summer bioinformatics interns on research projects and poster presentation skills (2016).
- Conducted multi-omic analyses (RNA-Seq, metagenomics) to identify microbial signatures associated with disease states.

Biostatistics Research Mentor / Teacher

Aug 2012 – Dec 2014

Universidad Metropolitana — Ana G. Méndez University System, San Juan, PR

- Began teaching biostatistics at age 19–20 — with students' parents consistently remarking on her command of the material at such a young age.
- Developed the teaching curriculum for the biostatistics research group, including teaching materials, slides, data samples, and quizzes.
- Authored and submitted research proposals for more than 15 student cohort projects in data science and statistics applied to health, genomics, and biological data.
- Taught cohorts of 15+ students — covering scientific method, coding in R and Python, and research ethics — fostering an inclusive, curiosity-driven learning environment; graded homework and projects and gave presentation feedback.

Research Intern & Research Assistant, Statistical Genetics

Jun 2013 – Feb 2015

Section of Statistical Genetics, Department of Public Health, University of Alabama at Birmingham, AL · Mentors: Dr. Ana I. Vazquez, Dr. Hemant K. Tiwari

- Research Intern in Summer 2013, then Research Assistant from Summer 2014, continuing remotely through February 2015 to finalize the analysis and prepare the manuscript for a first-author, peer-reviewed paper in *Frontiers in Genetics* (2015), published while still an undergraduate.
- Estimated genetic variances of obesity-related genes using Bayesian G-BLUP models on a 5,000-patient genomic and physical dataset (TIGER study), achieving a 2.8% accuracy improvement over published benchmarks.
- Built ML models (logistic regression, CNN, SVM) for Type 2 Diabetes risk prediction from Framingham Heart Study clinical and genetic data, using 65 PCA-reduced SNPs, detecting ~15 possible SNP biomarkers and yielding 80% AUC via 10-fold cross-validation; published in *Frontiers in Genetics* (2015).
- Communicated research advances to project stakeholders through oral presentations and written reports.

- Presented results at the UAB Research Expo (2013, third place for Best Poster in Public Health; 2014), the 71st Joint Annual NIS & BKX Meeting (2014), the Undergraduate Research Symposium in San Juan (2013, 2014) and UMARCH at the University of Puerto Rico, Mayagüez (2015).

Senior Data Analytics Tutor (Remote) & Data Analytics Teaching Assistant

May 2019 – May 2020

Trilogy Education

- Tutored over 200 students remotely, mentoring and guiding them through struggles in Python programming and machine learning/statistical methodology, resulting in measurable improvement in comprehension.
- As Teaching Assistant in Sacramento, CA (May–Jul 2019), provided in-person teaching support for a class of 20+ students: graded homework, gave feedback on data projects, and taught modules in statistics and hypothesis testing.
- Reviewed lessons and curriculum and stayed current on the latest data analytics tools; taught Python, pandas, matplotlib, HTML5, CSS3, JavaScript, SQL, MongoDB, machine learning, Tableau, Excel, and web scraping.

All-Levels Mathematics Tutor (Freelance)

May 2019 – May 2020

Independent, California

- Tutored 6 students from different backgrounds across multiple weekly sessions in Trigonometry, Geometry, Algebra, and Statistics.
- Supported students with homework, projects, and standardized test prep; provided guidance on study habits and coping strategies for test anxiety.

Lead Cashier

Jun 2010 – May 2015

Plaza San Miguel Grocery Market, San Juan, PR

- Delivered high-quality customer service across nearly five years, handling high-volume transactions accurately and efficiently while completing her bachelor's degree.
- Managed daily cash-flow accounting, inventory oversight, stocking, and product presentation to store standards.
- Assisted management in building employee work schedules to ensure adequate staffing and efficient operations.
- Supported team members during peak hours and contributed to a positive, collaborative work environment; made product recommendations and resolved customer inquiries directly.

UNDERGRADUATE RESEARCH EXPERIENCE

Undergraduate Researcher, Biomathematics

Aug 2014 – Feb 2015

Student Research Development Center, Universidad Metropolitana, San Juan, PR • Mentor: Dr. Luis F. De La Torre

- Population Dynamics of the Green Iguana in Puerto Rico: a Pest Control Method. Built discrete population models and Vensim simulations to estimate population growth and evaluate control strategies; presented at the 2015 Joint Mathematics Meetings (San Antonio, TX).

Undergraduate Intern, Biostatistics

Summer 2012

Iowa Summer Institute in Biostatistics (ISIB), University of Iowa, Iowa City, IA • Mentors: Dr. Jeffrey Dawson, Dr. Gideon Zamba

- Using Actigraphy Watches to Measure Sleep Activity in Subjects with Obstructive Sleep Apnea. Cleaned and formatted longitudinal actigraphy data, built longitudinal visualizations in R, and applied Wilcoxon rank-sum tests and ANOVA.
- Presented at the ISIB Symposium (2012), AGMUS Research Symposium (2012, Best Poster in Biostatistics), SACNAS National Conference (2012), Richard Tapia Conference (2013) and MGE@MSA/WAESO Conference (2013, Honorable Mention).

Undergraduate Researcher, Stream Ecology & Statistics

Jun 2011 – Aug 2011

Vermont EPSCoR, Saint Michael's College, Colchester, VT • Mentor: Dr. Declan McCabe

- Community Productivity with Sand Exposure in Urban and Forested Streams and the Effect of Fine Sediments on Grazers and Scrapers. Designed stream sampling sites, collected and taxonomically identified macroinvertebrates, and used ANOVA to compare abundances across urban and forested streams.
- Presented at the AGMUS Research Symposium (2011), SACNAS National Conference (2011, Best Poster in Applied Mathematics) and the 69th Joint Annual NIS & BKX Meeting (2012).

ADDITIONAL TEACHING, LEADERSHIP & SERVICE

- Mentor Assistant for Bioinformatics Interns, S Program, Global Security, Lawrence Livermore National Laboratory (Summer 2016).
- Mathematics Tutor, Learning Zone Center, Universidad Metropolitana (Fall 2013); Calculus I Tutor for pre-college students (Fall 2013).
- Judge, AGMUS Pre-College Symposium (Winter 2012 and Spring 2013).
- Vice-President, American Mathematical Society student chapter, Universidad Metropolitana (2011–2012).

- Speaker, 2016 EmpowHer Institute Maker Awards; volunteer, KDD 2016 Conference; mentor to high-school students at My Brother's Keeper Day, LLNL; Raspberry Pi booth attendant, BET Experience at LA Live (June 2016).

EDUCATION & CERTIFICATIONS

M.S. Statistics — University of California, Davis, 2016–2018 (completed while employed full time as a Data Scientist at Lawrence Livermore National Laboratory)

B.S. Applied Mathematics — Universidad Metropolitana, Puerto Rico, 2010–2015

Google Project Management Professional Certificate, 2024

Leland Product Management Program (competitive grant recipient), 2026

DeepLearning.AI — Deep Learning Specialization

IBM SQL Databases Professional Certificate

Human Research – Biomedical (IRB-01) and Responsible Conduct of Research (RCR) certifications

AWARDS & RECOGNITION

- 5+ poster presentation awards in life sciences and data science research.
- Best Poster Presentation in Computation — Lawrence Livermore National Laboratory Symposium (metagenomics contaminant profiling, 1000 Genomes Project).
- GEM Fellowship, Full Fellow (LLNL – UC Davis), 2016–2017.
- NSF Research Training Group Grant, 2016–2017.
- Travel Fellowship for the BPDW Workshop and KDD Conference, 2016 (awarded to ~10% of applicants).
- UAB Research Expo, third place for Best Poster Presentation in Public Health, 2013.
- MGE@MSA/WAESO Research Conference, Honorable Mention for Poster Presentation, 2013.
- AGMUS Research Symposium, Best Poster Presentation in Biostatistics, 2012.
- SACNAS National Conference, Best Poster Presentation in Applied Mathematics, 2011.
- NSF Bio-Mathematics Scholarship (2011–2014); Mathematics Alliance Scholar (2012–2014); AGMUS Bio-Mathematics Excellence List (Spring 2011); Model Institute of Excellence (MIE) Award (2011).
- Travel awards: SACNAS National Conference (2011, 2012), BKX 69th Annual Conference, MGE@MSA/WAESO Conference, Richard Tapia Conference (2013), 71st Joint Annual NIS & BKX Conference, Joint Mathematics Meetings (2015) and Field of Dreams (2014).

PATENT & SELECTED PUBLICATIONS

- Johansen, A., Bruns, S. C., Lebrón-Aldea, D., & Chapman, K. B. (2026). Methods for Obtaining RNA Secondary Structure of LNP-Encapsulated RNA. U.S. Patent Application Publication No. 2026/0250666 (PCT/US2023/067794), published Aug. 27, 2026.
- Lebrón-Aldea, D., Dhurandhar, E. J., Pérez-Rodríguez, P., Klimentidis, Y. C., Tiwari, H. K., & Vazquez, A. I. (2015). Integrated genomic and BMI analysis for type 2 diabetes risk assessment. *Frontiers in Genetics*, 6, 75. (First author; research completed as an undergraduate.)
- Thissen, J.B., ..., Lebrón-Aldea, D. et al. (2018). Novel torque teno virus 7 variant in Kawasaki disease patients. *PLOS ONE* 13(12): e0209683.
- Sichtig, H., ..., Lebrón-Aldea, D. et al. (2018). FDA-ARGOS: Quality-Controlled Genome Database for Infectious Disease Diagnostics. *Nature Communications*.